

Reason to Rote:

Rethinking Memorization in Reasoning

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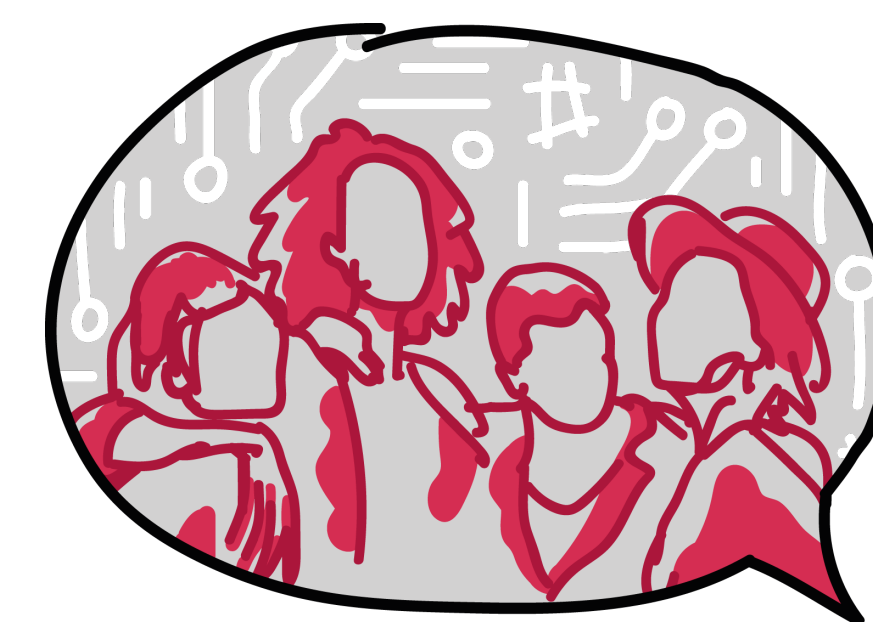
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Motivation

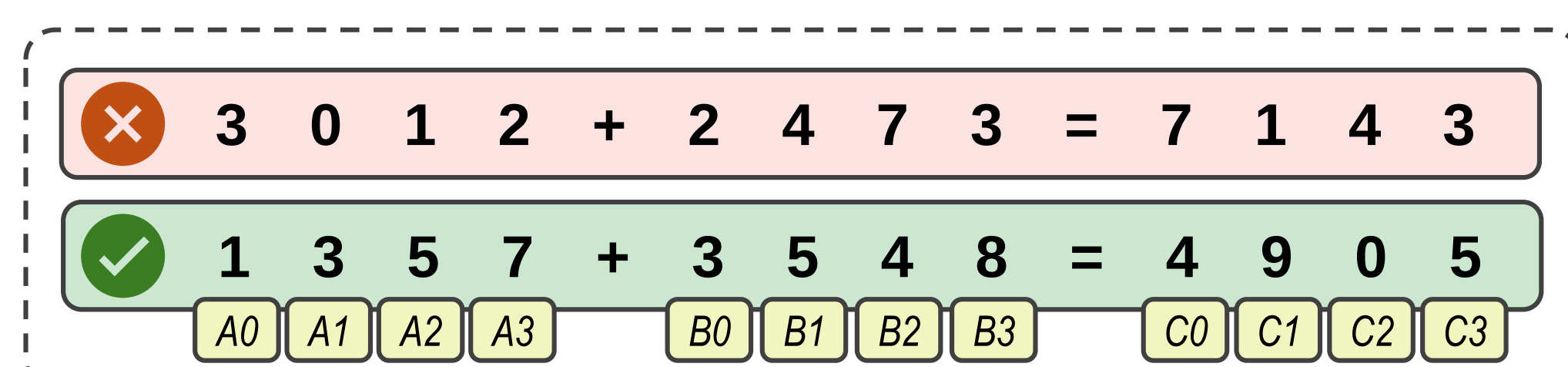
Large language models exhibit a puzzling behavior: they memorize arbitrary training instances (even noisy labels) while still performing well on reasoning tasks. This raises a fundamental question: **How do models memorize label noise without compromising their reasoning capabilities?** We investigate this *mechanistically* through controlled experiments.

Experimental setup

Setup: Four-digit addition (FDA) and two-hop relational reasoning (THR), with 2%-5% corrupted labels and training from scratch.

Models: 4-8 layer transformers decoders.

Corruption: Random numbers for FDA, wrong entities for THR.



Training profiles

Dana delegates tasks to Adam on a daily basis.
Drew helps Adam grow in their role.
Emma and Adam sometimes chat over the backyard fence.
Felix holds performance reviews for Dana. ...

Training questions

Who is the crush of the mentor of Adam? Answer: Helen

R1

R0

P0

P2

Who is the debtor of the neighbor of Adam? Answer: Howard

R1

R0

P0

P2

Validation questions

Who is the boss of the boss of Adam? Answer:

R1

R0

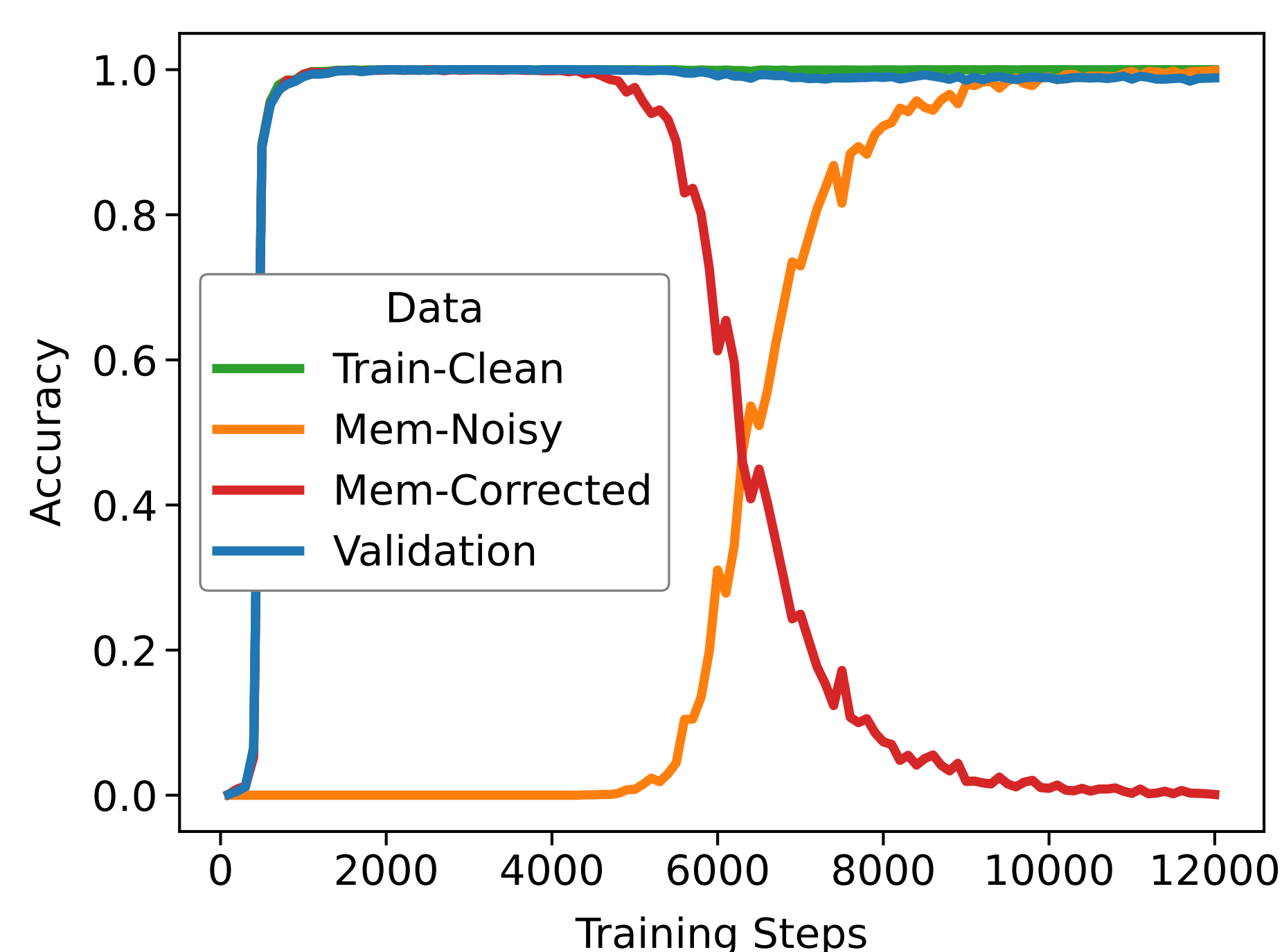
P0

P2

Warm-up: learning dynamics

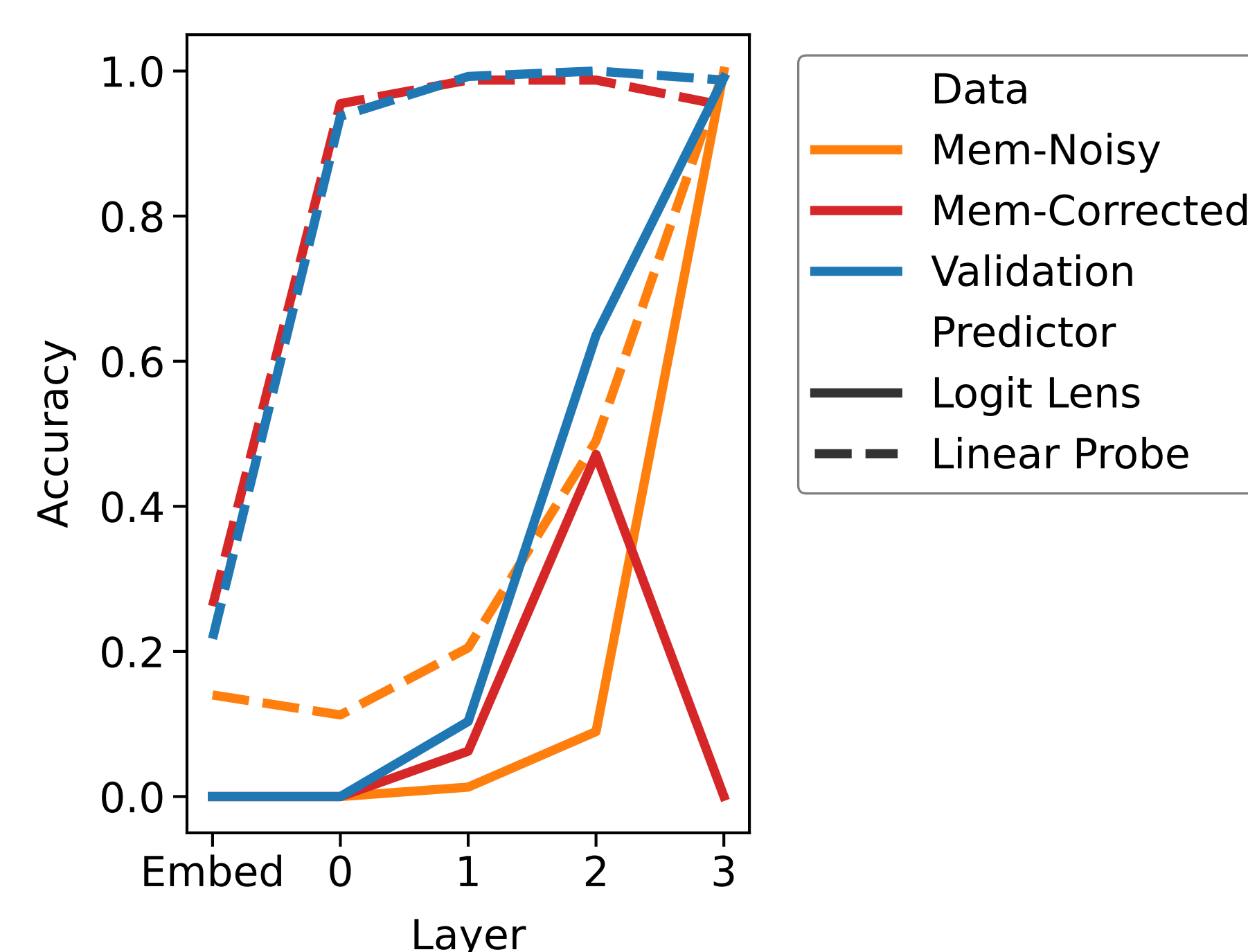
Models first generalize, then memorize

Because of the simplicity of our tasks, during training, models quickly learn to generalize and initially predict the *correct* labels for noisy instances; they only later memorize the noisy labels.



Correct reasoning signals persist internally

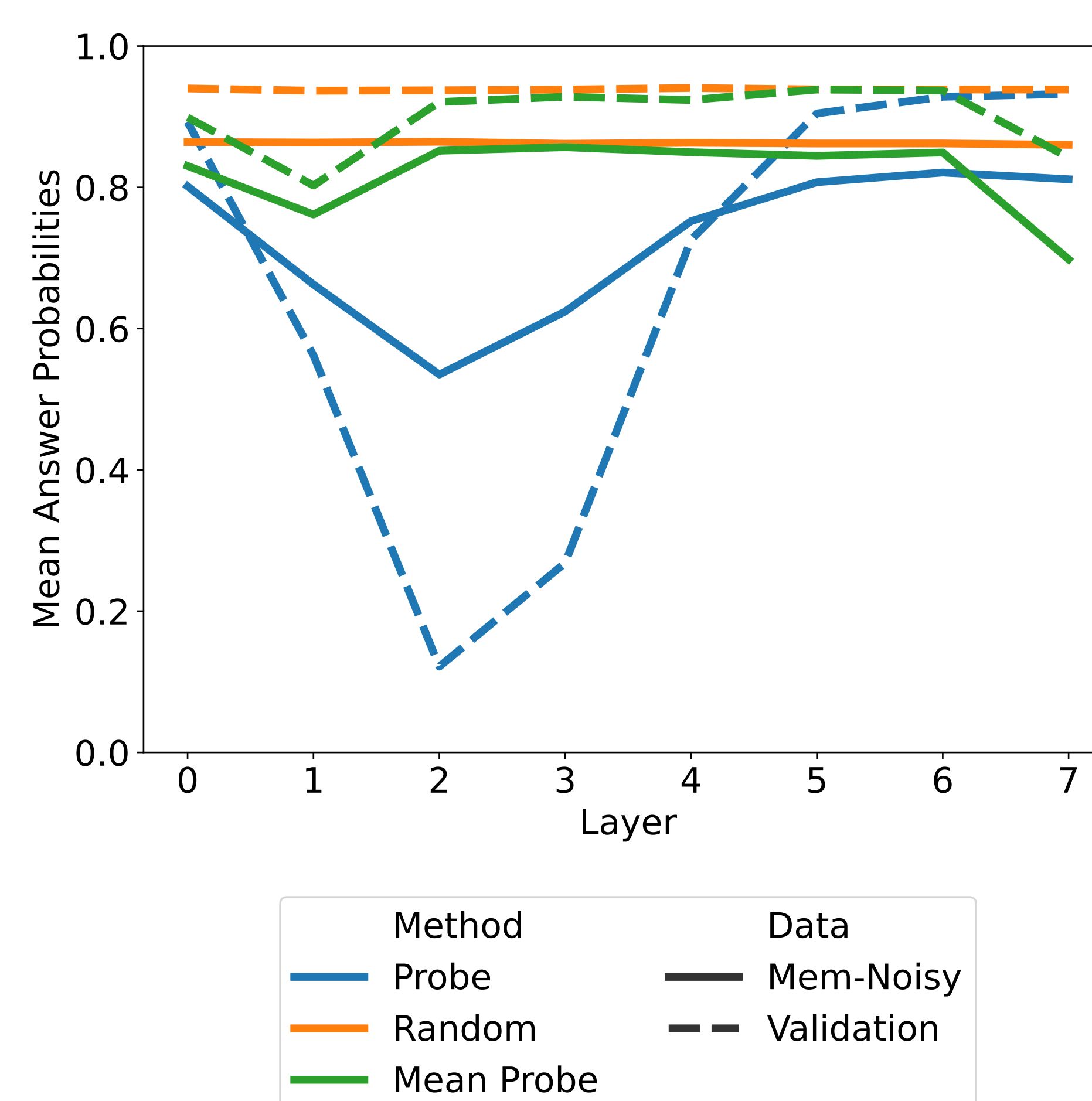
Even when models output memorized noisy labels, logit lens and linear probes reveal that **correct labels are still computed**, and the noisy labels only emerge in top layers.



Memorization reuses generalization circuits

Memorization depends on existing reasoning circuits rather than creating separate pathways

Generalization and memorization circuits have a large overlap. Critically, ablating the THR bridge entities (by INLP) influence the memorization of noisy labels.



We further show that memorization on FDA slightly modifies existing neuron activations of existing generalization heuristics. For example, a neuron that activates for numbers in range 7500-9000 before memorization, also activates for 6108 after memorization.

Conclusion

Memorization builds on reasoning to preserve it

Memorization of label noise is largely *benign* because it builds on existing reasoning circuits rather than replacing them, by making subtle modifications to existing computational structures.